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[> # Электронная подпись с хэшированием
[
[0 (1)
[
[> with(numtheory) :
[> with(RandomTools) :
[> hash := (x) → x mod 5;
[ hash := x ↦ x mod 5 (2)
[
[> p1 := nextprime(Generate(integer(range=2..10^6)));
[ p1 := 113513 (3)
[
[> p2 := nextprime(Generate(integer(range=2..10^6)));
[ p2 := 958213 (4)
[
[> M := p1 · p2;
[ M := 108769632269 (5)
[
[> ifactor(M);
[ (113513) (958213) (6)
[
[> d := Generate(integer(range=2..10^4));
[ d := 8059 (7)
[
[> igcd(d, phi(M));
[ 1 (8)
[
[> d · e = 1 mod phi(M);
[ 754309967372641 = 1 (9)
[
[> e := d &^ (phi(phi(M)) - 1) mod (phi(M));
[ e := 93598457299 (10)
[
[> x := Generate(integer(range=10^4..10^15)) mod M; # Сообщение
[ x := 16300367058 (11)
[
[> s := hash(x) &^ d mod M;
[ s := 95734047032 (12)
[
[> x1 := s &^ e mod M;
[ x1 := 3 (13)
[
[> hash(x) - x1; # Проверка
[ 0 (14)
[>

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